time to find such errors. Meanwhile, Perry et al. (2022) find that humans miss security vulnerabilities introduced by LLM code assistants. Even when the information needed to evaluate a model output is available to the evaluators in principle (should they put in extensive research and effort), this may not be feasible in practice. Bowman et al. (2022) formulate tasks on which nonexpert humans struggle to grade answers to questions accurately and argue that human feedback alone will not be sufficient to exercise scalable oversight for superhuman AI systems.

Fundamental: Humans can be misled, so their evaluations can be gamed. Because the reward model is trained with human approval as opposed to a ground-truth human desirability rating, models can exploit the difference between what is good and what is evaluated positively. Language models can imitate the persuasive and manipulative tactics of humans (Bai, 2023; Vincent, 2023; Griffin et al., 2023). In particular, language models trained with RLHF can sound confident even when they are incorrect (Snoswell and Burgess, 2022) which can lead humans to provide more positive feedback (Bowman et al., 2022). These incentives to mislead also connect to broader worries about manipulation (Kenton et al., 2021; Carroll et al., 2023; Everitt et al., 2021). In addition to sounding confident, RLHF can contribute to sycophancy (Perez et al., 2022b), or "gaslighting" of humans (Vincent, 2023). Misleading behavior will actively be incentivized by RLHF when humans can be tricked into mistakenly providing positive feedback (Carroll et al., 2023; Steinhardt, 2023).

3.1.3 Data Quality

Obtaining representative and helpful data is an open technical problem.

Tractable: Data collection can introduce harmful biases. Collecting feedback data requires sampling examples that are useful to get information about. Ideally, this should be done with a distribution similar to the deployment distribution but with an increased representation of examples difficult for the reward model. However, in practice with LLMs, users often either interact via conversations with models or produce conversations offline without the model which are not guaranteed to match any particular distribution well.

Fundamental: There is an inherent cost/quality tradeoff when collecting human feedback. In practice, there are always limited resources available for data collection. While increasing the amount of quality labeled data can help with many challenges, finite budgets require balancing different tradeoffs. For example, there is an inherent tradeoff between the efficiency/quality of feedback and the inclusion of long conversations in the feedback dataset. Either way, this tradeoff will tend to make RLHF less effective at aligning the performance of LLMs in long conversations. Helpful approaches for improving data quality have been to obtain samples that are diverse (Zhou et al., 2023), adversarial (Ziegler et al., 2022), and which the reward model is uncertain about (Christiano et al., 2017). However, active learning techniques in deep learning rely on heuristics for prediction confidence which can be unreliable (Gleave and Irving, 2022). Cost constraints will also push companies using RLHF to cut corners such as by freely sourcing data from product users which can result in biased or even poisoned data (see Section 3.1.1). Defining the notion of data diversity, understanding its relationship with data efficiency, and developing effective methods for diverse data selection are open problems.

3.1.4 Limitations of Feedback Types

Fundamental: RLHF suffers from a tradeoff between the richness and efficiency of feedback types. Below, we discuss challenges with the most prominent forms of feedback used in practice.

Comparison-based feedback: The most common type of feedback used with RLHF is binary preferences between pairs of examples (Christiano et al., 2017) though k-wise rankings (Brown et al., 2019; 2020; Zhu et al., 2023; Myers et al., 2021) or best-of-k queries (Biyik et al., 2019) can be used as well. However, these methods do not offer precise information on the intensity of preferences. A learned preference ordering can fail to converge to the true one when the desirability of examples depends on noise or unmodeled, contextual details not contained in the observations (e.g., randomness in a human's feedback or differences between evaluators (Myers et al., 2021)). Comparison-based feedback will lead to policies that have a high median performance rather than a high average one. Consider a simple example in which actions of type A are always recognized to be of value 1 to an evaluator, while actions type B are recognized to have value 10 on

40% of examples but are overlooked and concluded to have value 0 on 60%. Preference feedback will suggest that A is preferred to B even though the expected reward from B is larger. See also Section 3.2.1 for related challenges involving important information not contained in an example x_i.

Scalar feedback: Obtaining scalar feedback addresses some problems of comparison-based feedback – it is significantly more expressive (Wilde et al., 2022). However, scalar rewards from humans can be poorly calibrated. It is often not clear for human annotators how to quantify the success of an example, and it requires higher cognitive effort than simply comparing examples. Scalar feedback is more susceptible to inconsistency between annotators and suffers from bias due to the order in which examples are presented (Yannakakis and Hallam, 2011). A combination of comparison and scalar feedback where the annotators indicated the intensity of a preference using a slider bar was demonstrated by Wilde et al. (2022), but it requires more sophisticated and annotator-specific human response models. Attempting to discretize this form of feedback using a Likert scale (a range of discrete ratings; e.g., very bad, bad, ok, good, very good) simplifies the process of feedback collection (Knox and Stone, 2008; MacGlashan et al., 2017; Arumugam et al., 2019). However, the resulting learned preference ranking can be the opposite of the true one when assumptions commonly made in practice are violated (Ethayarajh and Jurafsky, 2022).

Label feedback: Sometimes, humans can provide feedback in the form of classifying examples. Label selection can be low-effort, but often suffers from *choice set misspecification* (Freedman et al., 2021; Guerdan et al., 2023; Casper et al., 2023b) when the given options don't fully encompass the labels needed to properly describe the data. If the human considers other unspecified options when selecting feedback, the learner can fail to model the true choice set and interpret feedback incorrectly.

Correction feedback: Feedback can come in the form of corrective demonstrations or adjustments that improve on an example from the model. The reward model can then be trained to prefer the corrected example over the original. In robotics, correction-based feedback has been used for improving policies (Li et al., 2021; Losey et al., 2022; Bajcsy et al., 2018) and plans (Sharma et al., 2022). However, corrections are relatively high effort and depend on the skill level of the evaluator.

Language feedback: Using language, humans can convey a large amount of information per evaluation, reducing ambiguity and goal misspecification. Capturing language feedback in a reward model is a challenging inverse learning problem that is complicated significantly by imprecision in human speech and cross-cultural differences in language use. A body of work on using language feedback for reward inference and shaping might lessen this challenge (Fu et al., 2019; Goyal et al., 2019; Sumers et al., 2021; Zhou and Small, 2021; Lin et al., 2022; Yu et al., 2023), but thus far, these techniques have not been applied to LLMs. See also Section 4.2 for a discussion of related methods that use human language feedback for training LLM policies *without* using a reward model (which excludes them from our definition of RLHF).

3.2 Challenges with the Reward Model

Here, we discuss challenges resulting from misspecification, misgeneralization, reward hacking, and evaluating the reward model. Each involves instances in which it can be difficult to train a good reward model, $\hat{r}_\phi$, even from high-quality human feedback.

3.2.1 Problem Misspecification

The standard approach to fitting a reward model to represent human values is a doubly-misspecified problem.

Fundamental: An individual human's values are difficult to represent with a reward function. Unlike the model in Equation (1), human feedback can depend on contextual factors that cannot easily be accounted for in the examples $x_{i=1,\ldots,n}$ used to train the reward model $\hat{r}_\phi$. Humans possess a range of intricate and context-dependent preferences that evolve over time and are difficult to model accurately. Models of human goals based on incorrect assumptions about human decision-making can impair reward inference (Hong et al., 2022). Even modeling human preferences with a reward at all, implicitly accepting the reward hypothesis (Silver et al., 2021), might be unwarranted (Skalse and Abate, 2022b; Bowling et al., 2023; Vamplew et al., 2022; Bobu et al., 2023). A number of studies have examined incorrect assumptions in various aspects of human models, such as their use of regret (Knox et al., 2022), the hypothesis space

of reward models (Bobu et al., 2020; Biyik et al., 2020), and pedagogic behavior (Milli and Dragan, 2020). Skalse and Abate (2022a) formally study the effect of inverse reinforcement learning with a misspecified Boltzmann model, which is also common (Jeon et al., 2020). Most work in RLHF does not take into account personality and context-dependence of human preferences (Milano et al., 2021; Lindner and El-Assady, 2022), and Zhao et al. (2016) prove a mixture of reward functions cannot be identified from binary preferences without additional context. Different models for the human can also be better or worse for learnability (Knox et al., 2022). In particular, modeling human irrationalities can make reward learning difficult (Nguyen et al., 2017; Mindermann and Armstrong, 2018; Shah et al., 2019), leading to a trade-off between efficiency and accuracy. Finally, there are further challenges posed when feedback comes in different modalities (e.g., demonstrations and preferences). Jeon et al. (2020) and Bıyık et al. (2022) propose ways of combining different types of information about human goals, but these approaches are sensitive to modeling assumptions about the human.

Fundamental: **A single reward function cannot represent a diverse society of humans.** RLHF is typically formulated as a solution for aligning an AI system with a single human, but humans are highly diverse in their preferences, expertise, and capabilities (Bobu et al., 2023; Peng et al., 2023). Evaluators often disagree: Stiennon et al. (2020), Ouyang et al. (2022), and Bai et al. (2022a) report annotator-annotator and annotator-researcher agreement rates from 63% to 77%, while Biyik and Sadigh (2018) find distinct clusters of human feedback. Attempting to condense feedback from a variety of humans into a single reward model without taking these differences into account is thus a fundamentally misspecified problem. Moreover, current techniques model differences among evaluators as noise rather than potentially important sources of disagreement (Baumler et al., 2023) (see Equation (1)). As a result, when preferences differ, the majority wins, potentially disadvantaging under-represented groups (Prabhakaran et al., 2021; Feffer et al., 2023; Kirk et al., 2023).

3.2.2 Reward Misgeneralization and Hacking

Reward models tend to be imperfect, and imperfection in reward models leads to reward hacking.

Fundamental: **Reward models can misgeneralize to be poor reward proxies, even from correctly-labeled training data.** There can exist many ways to fit the human feedback dataset $\mathcal{D} = \{(x, y)_{i=1,\ldots,n}\}$, even in the limit of infinite training data (Skalse et al., 2023). Reward models can compute reward using unexpected, possibly contingent features of the environment (Michaud et al., 2020) and are prone to causal confusion and poor out-of-distribution generalization (Tien et al., 2023). Reward learning algorithms can even produce reward models that fail to train new agents from scratch in various settings, raising concerns about their reliability as signals for policy learning (McKinney et al., 2023).

Fundamental: **Optimizing for an imperfect reward proxy leads to reward hacking.** Reward models can differ from humans due to misspecification (Section 3.2.1) and misgeneralization (Section 3.2.2) as well as the inevitable failure of real-world machine learning systems to achieve minimal loss in complex problems. Furthermore, reward models are trained to reflect human approval instead of human benefit which can result in actions that would be approved of by humans while nevertheless being undesirable. Applying strong optimization pressure for an imperfect proxy measure for a goal tends to cause poor performance on the underlying target goal (Hoskin, 1996; Manheim and Garrabrant, 2018; Gao et al., 2022). For example, without regularization penalizing the KL divergence between a base model and the finetuned model, LLMs undergoing RL often learn to output nonsensical text (Ziegler et al., 2019; Stiennon et al., 2020). This type of problem is known as "reward hacking", and has been observed in AI systems, including those trained with RLHF (Skalse et al., 2022; Krakovna et al., 2020). Skalse et al. (2022) show that unhackable proxies are very rare in complex environments, and Zhuang and Hadfield-Menell (2020) prove under mild conditions that reward hacking should be expected by default. Using a suite of environments Pan et al. (2022) find that reward hacking also becomes more likely as an agent's raw capabilities increase.

3.2.3 Evaluating Reward Models

Tractable: **Evaluating reward models is difficult and expensive.** When the true reward function is known, several methods can be used to judge the quality of the learned reward model (Gleave et al.,